

FIRMWARE FLOWCHART

ESP32 main loop — sensor acquisition to cloud sync

DOC HA-2024-SW-01
REV 01
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START



PROC-01 · INITIALISATION

Initialize Components

ESP32 · RGB LEDs · Buzzer · OLED Display · MQ-135 Sensor · SGP30 Sensor



PROC-02 · NETWORK

Connect to Wi-Fi

Attempt connection · Retry with backoff on failure · Confirm SSID handshake



PROC-03 · VOICE INTEGRATION

Setup Alexa via Sinric Pro

Register device · Authenticate API key · Enable Sinric Pro skill handlers

↓ MAIN LOOP — REPEATS EVERY 1 SECOND



PROC-04 · SENSOR ACQUISITION

Read MQ-135 + SGP30 Sensor Data

MQ-135 → CO₂ concentration (ppm) · SGP30 → eCO₂ + TVOC (ppb)

CLASSIFY
AIR QUALITY?

GOOD

MODERATE

BAD

